

	Lower intercept with vertical axis, smaller gradient	[1]
5(a)(i)	Voltmeter reading, $V = \frac{5.00}{5.00 + 1.25}(9.00) = 7.20 \text{ V}$	[1] ans
(a)(ii)	The diagram shows a DC circuit. On the left, a battery is represented by two parallel horizontal lines of unequal length. A vertical wire connects the top of the battery to a junction. From this junction, the circuit splits into two parallel branches. The upper branch contains an LDR, represented by a rectangle inside a circle, with two arrows pointing towards it. The lower branch contains a lamp, represented by a circle with a cross inside. Both branches rejoin at a second junction, which is connected to the bottom of the battery by a vertical wire.	[1] LDR in series with the resistor [1] LDR // with the lamp

(b)(i)	Potential difference across $100\ \Omega$ resistor = potential difference across balanced length 400 mm. Potential difference across $100\ \Omega + R\ \Omega$ resistor = potential difference across balanced length 588 mm. $\Rightarrow \frac{400}{588} = \frac{100}{100 + R}$ $R = 47\ \Omega$	[1] sub [1] ans
(b)(ii)	Terminal potential for the 6.00 V source $= \frac{147}{147 + r}(6.00)$ Potential difference across balanced length of 588 mm $= \frac{588}{1000}(9.00)$ $\Rightarrow \frac{588}{1000}(9.00) = \frac{147}{147 + r}(6.00)$ $r \approx 19.7\ \Omega$	[1] sub [1] ans
6(a)	Work function energy refers to the minimum amount of energy	[1]